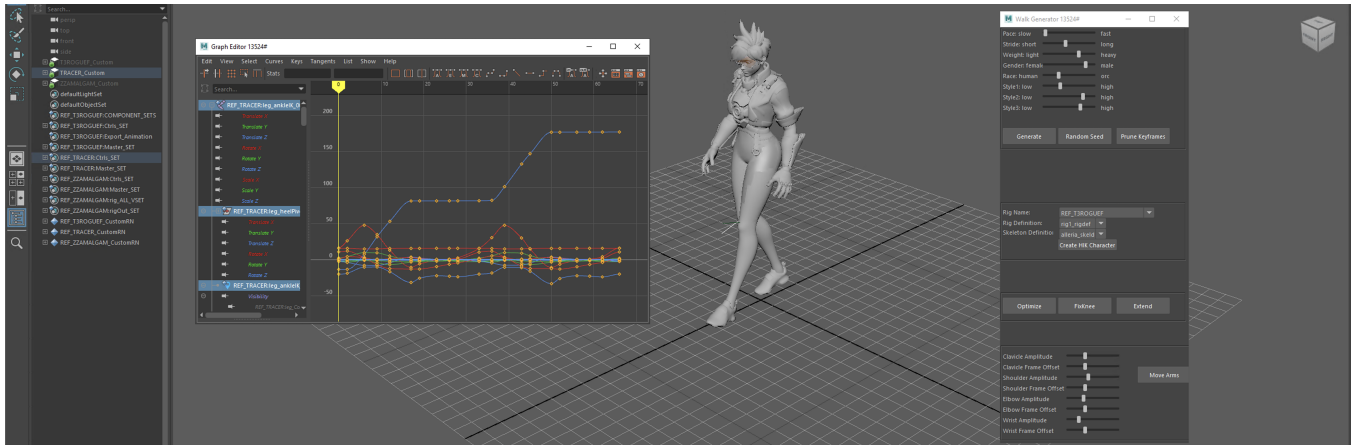


Artistically Directable Walk Generation

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ABSTRACT

We present a framework for artistically directable walk generation. A generative network is trained using a motion capture dataset and a manually animated collection of walks. To accommodate an animator's workflow, each walk is presented as a sequence of key poses. The generative framework allows to specify a set of traits including gender, stride, velocity and weight. A generated walk is designed to be the starting point when blocking an animation: an animator can introduce new keys on the controls.

KEYWORDS

character animation, artistically directable animation, machine learning, generative networks

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1 OVERVIEW

Walk cycles are always needed for hero and background animations but are time consuming processes with manual editing/framing of keys. A typical workflow involves blocking a walk with a sequence of key poses, which is further enhanced by additional keys on

select controls in order to convey an animator's intent. In this paper we propose a generative framework that allows to simplify the animation of walk cycles. We accomplished this by training on a generic rig with latent space parameters such as stride, speed, weight, gender etc. The generated walk cycle produces sparse key frames on a generic rig, which is further retargeted onto animator chosen specific rigs with post retargeting solvers to solve onto controls depending on various animator workflow styles.

2 RELATED WORK

The problem of generating keys was addressed by the research community in the previous years. In [Harvey et al. 2021], the authors synthesize animation using sparse keyframes as a starting point. A similar approach is investigated in [Zhou et al. 2020]. Starting with a dense motion capture dataset, these method produce dense keys on all frames and hence are hard to augment change in a workflow. In relation we take the animator workflow into account and precondition our training data resulting in scalable and artistically directable workflows.

3 METHOD

Any animation, including walk cycles normally proceeds by blocking. Based on Animators Survival Kit by Richard Williams (1a) a walk can be represented by a sequence of 5 key poses at bare minimum. The values and placement of these keys change depending on traits like speed, stride, weight, gender etc. While all animations are programmatically evaluated using bone/joint orientations, animators prefer to work with controls rather than joints. Therefore, our framework ultimately produces keys directly on the controls bypassing bones. Our system can be best explained in two parts - a training model/evaluation framework, and a post retargeting

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solver setup to solve the orientations onto specific rigs and controls depending on varying animator workflows.

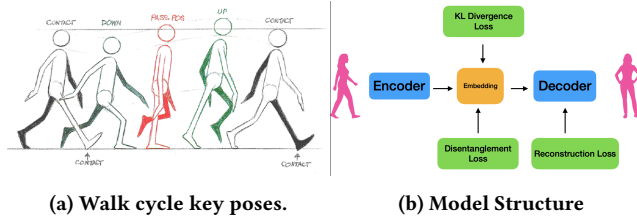


Figure 1

3.1 Training Model and Evaluation Framework

We start with manually animated walk datasets at our studio, along with a motion capture dataset [CMU 2003]. We have 320 walks, with roughly half of them being manually animated. For each, we extract the 5 base key poses 1a algorithmically. For example, a contact pose is described by the largest rotated X ankle in the heel joint. These five base poses for each walk are retargeted onto our generic rig and serve as the basis of our normalized training set. Each walk in the dataset is also annotated with the labels that an animator might want to have artistic control over e. g. weight, speed, gender, stride etc. We use a Variational Autoencoder which consists of an Encoder to compress our data into a latent space of size 8 and a Decoder (Generator) to create a new walk. See 1b for the model outline. Using ideas introduced in [Härkönen et al. 2020] and [Shoshan et al. 2021], the hidden space is disentangled by using a 2 layer feed forward neural network with ReLU activations. We add a loss term that reflects the ability of the 2 layer network to predict a latent variable using the rest of the latent variables. Since some of the latent variables contain direct physical meaning e. g. stride and speed, we put additional constraints on these by imposing that their softplus activations are positive before passing them over to the Generator. Additional Generator constraints can also be imposed by observing that the walk cycle is periodic: the last contact pose can be predicted as a reflection of the first contact pose 1a. Finally, most of the controls are modeled with a periodic function. For example, for an arm, we predict amplitude, phase shift and frequency. This helps reduce the parameter space since the dataset we use is small. See 1b for a depiction of loss terms incorporated into the model.

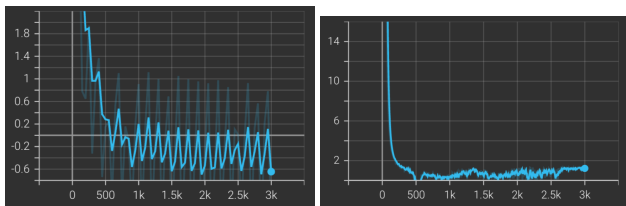


Figure 2

3.2 Retargeting and post solving setup

The sparse blocked key poses generated by latent slider inputs on the training rig are then retargeted onto a specific character

rig depending on the shot, with additional post solvers that allow for specific animator defined setups. For example, an animator is likely to prefer to work with heel when foot is stepping on the ground, and with peel control when the foot is lifting up. Since our model generates keys on specific controls on a sparse set of frames, animators can easily iterate by adding/augmenting values as needed.

4 EXPERIMENTS AND RESULTS

The training converges well, see 2a and 2b. A trained network generates naturally looking walks. The features attributed to specific labels are captured by the walk Generator. Some changes are subtle while others are distinctly visible. In particular, for a female walk, the amplitude in hips is larger than in the male walks. The heavier walks produce pelvis curve in translate Y that slowly builds up from Contact to Up pose and then swiftly plunges. We noticed that the constraints that describe a walk, for instance, the constant speed of center of mass, and the lack of foot sliding, are better incorporated in the Decoder itself rather than the loss terms since this results in better convergence. The abundance of degrees of freedom hurts convergence. When the constraints are taken into account as l2 loss terms, for example, the generated walks exhibit unnatural movements.

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REFERENCES

- CMU. 2003. CMU mocap. (2003). <http://mocap.cs.cmu.edu/>
- Erik Härkönen, Aaron Hertzmann, Jaakko Lehtinen, and Sylvain Paris. 2020. GANSpace: Discovering Interpretable GAN Controls. In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, Hugo Larochelle, Marc'Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin (Eds.). <https://proceedings.neurips.cc/paper/2020/hash/6fe43269967adbb64ec6149852b5cc3e-Abstract.html>
- Félix G. Harvey, Mike Yurick, Derek Nowrouzezahrai, and Christopher J. Pal. 2021. Robust Motion In-betweening. *CoRR* abs/2102.04942 (2021). [arXiv:2102.04942](https://arxiv.org/abs/2102.04942)
- Alon Shoshan, Nadav Bhonker, Igor Kviatkovsky, and Gérard G. Medioni. 2021. GAN-Control: Explicitly Controllable GANs. *CoRR* abs/2101.02477 (2021). [arXiv:2101.02477](https://arxiv.org/abs/2101.02477)
- Yi Zhou, Jingwan Lu, Connelly Barnes, Jimei Yang, Sitao Xiang, and Hao Li. 2020. Generative Tweening: Long-term Inbetweening of 3D Human Motions. *CoRR* abs/2005.08891 (2020). [arXiv:2005.08891](https://arxiv.org/abs/2005.08891)